

JEE Main - 8 | JEE - 2023 (Gen 1 & 2)

Date: 18/01/2022

Maximum Marks: 300

Timing: 04:00 PM - 09:00 PM

Duration: 3.0 Hrs

General Instructions

1. The test is of **3 hours** duration and the maximum marks is **300**.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is an **integer** ranging from 0 to 99. (both inclusive)
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them.**
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) :

Roll Number :

OMR Bar Code Number :

Candidate's Signature : Invigilator's Signature

SYLLABUS

Physics : Liquids, Properties of Matter

Chemistry (Revised) : Stoichiometry II, IOC

Maths : Sequence & Series, Trigonometry, Complex Numbers, PnC, Functions, Parabola, Ellipse

PART - I : PHYSICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- A simple pendulum consists of a metal wire of coefficient of linear thermal expansion α attached to a heavy bob. The time period of this pendulum on a cold winter day with temperature 2°C is T_C , and its time period on a hot summer day with temperature 42°C is T_H . The ratio $\frac{T_C}{T_H}$ is close to:

(A) $1 + 20\alpha$ (B) $1 - 20\alpha$ (C) $1 + 40\alpha$ (D) $1 - 40\alpha$
- Let us assume that the Sun and another star A are perfect spheres and perfect black bodies. The radius of star A is four times the radius of the Sun and it radiates heat energy at exactly the same rate as the Sun. If the surface temperature of the Sun is 6000 K , the surface temperature (in Kelvin) of the star A is:

(A) 8484 (B) 4242 (C) 3000 (D) 2000
- Two cuboidal blocks made of different materials of thermal conductivity K and $3K$ are placed end to end between two heat reservoirs maintained at constant temperature T_A and T_B (such that $T_A > T_B$). The blocks have the same area of cross-section, but lengths L and $2L$ respectively. The temperature at the interface between the blocks is:

(A) $T = \frac{1}{7}(T_A + 6T_B)$

(B) $T = \frac{1}{7}(6T_A + T_B)$

(C) $T = \frac{1}{5}(2T_A + 3T_B)$

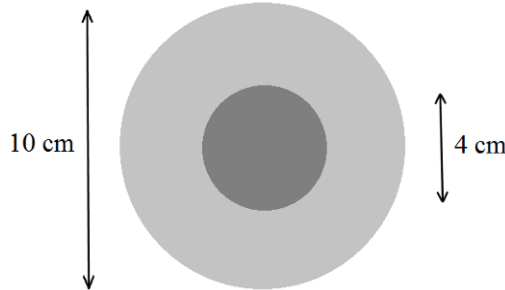
(D) $T = \frac{1}{5}(3T_A + 2T_B)$

	Insulation		
T_A	K	$3K$	T_B
	Insulation		
- If 50 g of ice at 0°C is mixed with 40 g of steam at 100°C in a calorimeter of water equivalent 200 g that was initially at 20°C , the final temperature (in $^\circ\text{C}$) is:

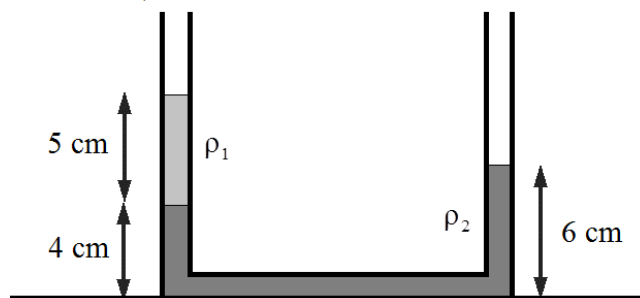
(Assume that the calorimeter is completely insulated from the surrounding so there is no heat loss, latent heat of fusion of ice = 80 cal/g , latent heat of condensation of steam = 540 cal/g , specific heat of water = 1 cal/g)

(A) 52.3 (B) 79.6 (C) 88.3 (D) 92.9

5. A sphere is composed of an inner “core” of diameter 4 cm made of Material A (density 3.0 g/cm^3) and an outer “cladding” of outer diameter 10 cm and inner diameter 4 cm made of Material B (density 0.5 g/cm^3). If the sphere is released into a vessel containing water (density 1.0 g/cm^3):

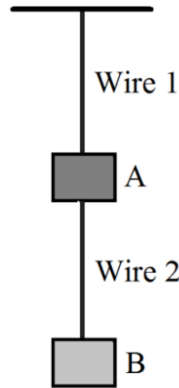


- (A) It floats with 83% of its volume submerged inside the water
 (B) It floats with 76% of its volume submerged inside the water
 (C) It floats with 66% of its volume submerged inside the water
 (D) It sinks
6. A spherical metal ball of radius r and density ρ , when dropped into a tall column of an incompressible liquid of density $\rho/2$ and coefficient of viscosity η_1 , has a terminal velocity v_1 , while a spherical metal ball of radius $2r$ and density ρ , when dropped into a tall column of an incompressible liquid of density $\rho/5$ and coefficient of viscosity η_2 , has a terminal velocity v_2 . The correct relation is:
- (A) $\frac{v_1}{v_2} = \frac{5 \eta_1}{8 \eta_2}$ (B) $\frac{v_1}{v_2} = \frac{5 \eta_2}{4 \eta_1}$ (C) $\frac{v_1}{v_2} = \frac{5 \eta_2}{16 \eta_1}$ (D) $\frac{v_1}{v_2} = \frac{5 \eta_2}{32 \eta_1}$
7. A narrow U-tube placed on a horizontal surface contains two liquids of density ρ_1 and ρ_2 as shown. The columns of the liquids are indicated in the figure. The ratio $\frac{\rho_1}{\rho_2}$ is equal to:

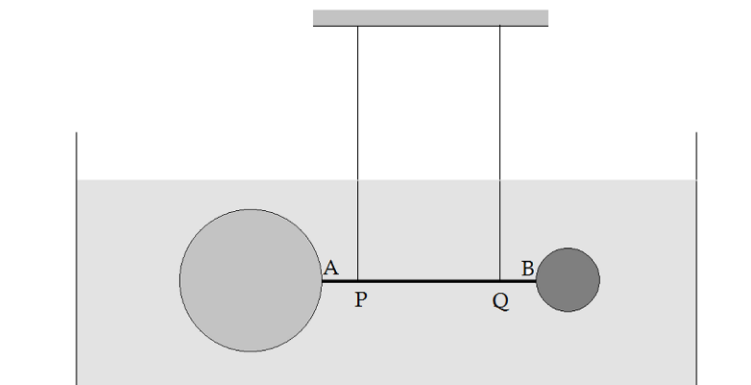


- (A) $\frac{1}{3}$ (B) $\frac{2}{5}$ (C) $\frac{1}{2}$ (D) $\frac{4}{5}$

8. Two blocks A and B of mass 4 kg and 2 kg respectively are suspended using two metallic wires as shown. The diameter of the wires is in the ratio $\frac{d_1}{d_2} = 1:2$, and their Young's modulus is in the ratio $\frac{Y_1}{Y_2} = 3:1$. The strain produced in them is in the ratio:



- (A) 2 : 1 (B) 4 : 9 (C) 2 : 3 (D) 4 : 1
9. A metallic sphere that can be assumed to be a black body is heated to temperature $2T$ and placed in an evacuated chamber which is maintained at temperature T . The initial rate of cooling of the sphere is R_1 , and its rate of cooling when its temperature has dropped to $3T/2$ is R_2 . The ratio R_1/R_2 is:
- (A) $\frac{64}{13}$ (B) $\frac{48}{13}$ (C) $\frac{36}{13}$ (D) $\frac{24}{13}$
10. Two uniform spheres of radius $2R$ and R , density 2ρ and 8ρ , and mass $2M$ and M respectively, are attached to a uniform rod of length $6R$, density 8ρ and mass $M/4$ at the points A and B such that the centres of both spheres are collinear with the rod. Now, the assembly is suspended by two massless strings, and submerged completely inside a liquid of density ρ filled in a container kept stationary on a horizontal surface as shown. The two strings are tied to the rod at points P and Q, such that $AP = QB = R$. The assembly is in equilibrium with the rod horizontal. The tension in the left and right strings is respectively:



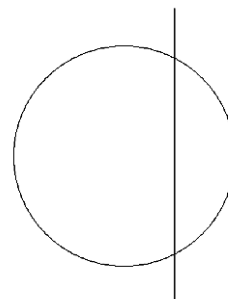
- (A) $\frac{91}{64}Mg, \frac{25}{64}Mg$ (B) $\frac{91}{64}Mg, \frac{43}{64}Mg$
 (C) $\frac{73}{64}Mg, \frac{43}{64}Mg$ (D) $\frac{83}{64}Mg, \frac{51}{64}Mg$

11. The specific heat capacity C (in J/kg K) of a substance varies with temperature T (in Kelvin) as: $C(T) = 4000 - 2T$. The heat required (in kJ) to raise the temperature of 0.1 kg of this substance from 300 K to 400 K is:

(A) 34 (B) 33 (C) 32 (D) 31

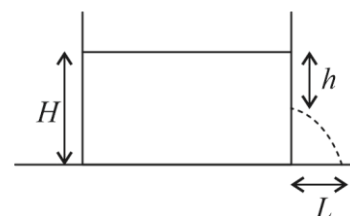
12. Let a spherical soap bubble be divided into two parts by an imaginary plane as shown. Then, the net force of surface tension applied on the right part by the left part is:

(A) zero
(B) directed perpendicular to the dividing plane, towards the centre of the bubble
(C) directed perpendicular to the dividing plane, away from the centre of the bubble
(D) directed parallel to the dividing plane

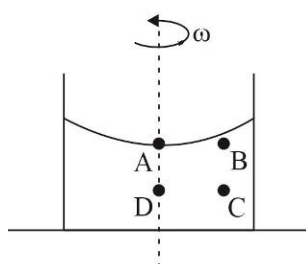


13. A liquid is filled in a container to a height H . A small hole is opened in the side of the container a height h below the surface of the liquid. Let the distance of the point where the stream of liquid leaving the container meets the ground from the bottom of the container be L . Then, if $h = 2H/5$, the ratio $\frac{L}{H}$ is equal to:

(A) $\frac{2}{\sqrt{15}}$ (B) $\frac{4}{\sqrt{15}}$ (C) $\frac{2\sqrt{6}}{5}$ (D) $\frac{4\sqrt{6}}{5}$



14. A cylindrical container filled with a liquid is being rotated about its central axis at a constant angular velocity ω . Four points A, B, C and D are chosen in the same plane such that ABCD is a square of side length a and AB is horizontal while BC is vertical. A and D lie on the axis of rotation. Let the pressure at A, B, C and D be denoted by P_A, P_B, P_C and P_D . Now, consider the following two statements:

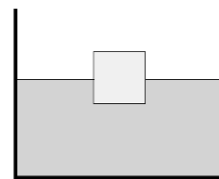


(I) $P_C < P_A$ for all values of ω (II) $P_B > P_D$ only if $\omega > \sqrt{\frac{2g}{a}}$

Which of these options is correct?

(A) Both (I) and (II) are correct (B) (I) is correct and (II) is incorrect
(C) (II) is correct and (I) is incorrect (D) Both (I) and (II) are incorrect

15. A solid cube floats in a liquid of density ρ with half of its volume submerged as shown. The side length of the cube is a , which is very small compared to the size of the container (the figure is not to scale). Now a liquid of density $\frac{\rho}{3}$ (immiscible with the liquid already in the container) is poured slowly into the container. The column of the liquid of density $\frac{\rho}{3}$ that must be poured so that the cube is fully submerged (with its top surface coincident with the surface of the poured liquid) is:



- (A) $\frac{a}{3}$ (B) $\frac{a}{2}$ (C) $\frac{2a}{3}$ (D) $\frac{3a}{4}$

16. The density of a metal at temperature T_1 is ρ_1 and its density at temperature T_2 is ρ_2 . The volume thermal expansion coefficient of the metal is given by:

- (A) $\gamma = \frac{3}{(T_2 - T_1)} \left(\frac{\rho_1}{\rho_2} - 1 \right)$ (B) $\gamma = \frac{3}{(T_2 - T_1)} \left(\frac{\rho_2}{\rho_1} - 1 \right)$
 (C) $\gamma = \frac{1}{(T_2 - T_1)} \left(\frac{\rho_1}{\rho_2} - 1 \right)$ (D) $\gamma = \frac{1}{(T_2 - T_1)} \left(\frac{\rho_2}{\rho_1} - 1 \right)$

17. When a capillary tube of small diameter is dipped vertically into a liquid such that the length of the capillary above the liquid surface is $H = 10$ cm, the liquid rises to a height 4 cm inside the capillary. The angle of contact of the liquid inside the capillary is 60° . Now, the capillary is slowly moved down vertically, and hence its length above the liquid surface, H , decreases. Which of these options is correct?

- (A) The liquid leaks from the upper end of the capillary as soon as H decreases below 4 cm
 (B) The liquid never leaks as H is decreased towards zero, the meniscus is still formed at the upper end of the capillary with the angle of contact less than 60°
 (C) The liquid never leaks as H is decreased towards zero, the meniscus is still formed at the upper end of the capillary with the angle of contact more than 60°
 (D) The liquid never leaks as H is decreased towards zero, the meniscus is still formed at the upper end of the capillary and the meniscus is horizontal

18. A sphere made of steel is initially at the surface of a lake of depth 300 m, where the ambient temperature is 20°C . If the sphere is dropped into the lake such that it sinks and settles at the bottom, where the temperature is 10°C , the fractional change in its volume is:

(Bulk modulus of steel = 150 GPa, Volume thermal expansion coefficient of steel = $3 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$)

(Assume that the density of water everywhere in the lake is 1000 kg/m^3 , and $g = 10 \text{ m/s}^2$)

- (A) $-(10^{-5})$ (B) $-(2 \times 10^{-5})$ (C) $-(5 \times 10^{-5})$ (D) $-(6 \times 10^{-5})$

19. A body at temperature 80.0°C is placed in a room maintained at a constant temperature 20.0°C . The body takes time t_1 to cool from 80.0°C to 79.9°C , and it takes time t_2 to cool from 40.0°C to 39.9°C .

Assuming that the cooling process follows Newton's law of cooling, the ratio $\frac{t_1}{t_2}$ is close to:

- (A) $\frac{1}{6}$ (B) $\frac{1}{4}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$

20. Two spherical water drops of radius R and $2R$ combine to form one spherical drop. If the surface tension of water is σ , the reduction in surface energy due to the combining is:
- (A) $8\pi\sigma(5 - 3^{4/3})R^2$ (B) $4\pi\sigma(5 - 3^{4/3})R^2$
- (C) $8\pi\sigma(10 - 3^{5/3})R^2$ (D) $4\pi\sigma(10 - 3^{5/3})R^2$

SPACE FOR ROUGH WORK

SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 99 (both inclusive).

21. A 10 g ice cube at 0°C is added to an insulated vessel containing 150 g water at 80°C . Assuming that the heat capacity of the vessel is negligible, the final temperature (in $^\circ\text{C}$) of the mixture is _____. (Latent heat of fusion of ice = 336 J/g, specific heat of water = 4.2 J/g)
22. When a longitudinal force is applied on a uniform rod of diameter d made of Material P, the strain produced is 0.01%. If the same magnitude of longitudinal force is applied on a uniform rod of diameter $2d$ of Material Q, the strain produced is 0.02%. The ratio of the Young's modulus of the materials, $\frac{Y_P}{Y_Q}$ is equal to _____.
23. A sphere is made of an alloy of Metal A (density 8 g/cm^3) and Metal B (density 6 g/cm^3). The sphere floats in mercury (density 13.6 g/cm^3) with half its volume submerged. The percentage of the total volume of the sphere that is occupied by metal A is _____.
24. An otherwise thermally insulated room can only exchange heat with the surroundings through a window with a glass pane of area 1.2 m^2 and thickness 0.5 cm. A heater placed inside the room supplies heat to the room at a constant rate 1920 W. On a day when the temperature outside the room is 14°C , the temperature inside the room (in $^\circ\text{C}$) a long time after the heater is switched on becomes _____. (Thermal conductivity of glass = $1.0\text{ W/m}^\circ\text{C}$)
25. A cylindrical tube with a frictionless piston ends in a small hole of area of cross-section $4 \times 10^{-6}\text{ m}^2$, which is very small as compared to the area of cross-section of the tube. The tube is filled with a liquid of density 10^4 kg/m^3 . The tube is fixed horizontally and the piston is pushed at a constant speed so that liquid comes out of the hole at a constant volumetric rate $Q\text{ m}^3/\text{s}$. No liquid leaks around the piston. The material of the tube can bear a maximum pressure $6 \times 10^5\text{ Pa}$. The maximum possible value of Q so that the tube does not break is $\text{_____} \times 10^{-5}$. (Atmospheric pressure = 10^5 Pa)



SPACE FOR ROUGH WORK

PART - II : CHEMISTRY**100 MARKS****SECTION-1**

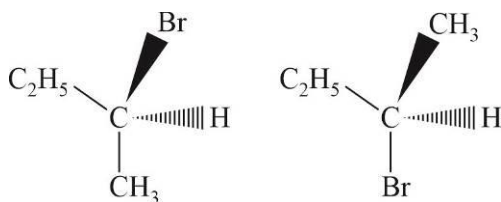
This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. An isomer of C_6H_{14} forms three monochloro derivatives. The isomer may be :

(Excluding stereo isomer)

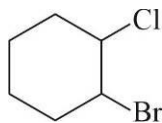
- (A) neo-pentane (B) n-hexane
(C) 2, 3-dimethylbutane (D) iso-hexane

2. What is the relationship between the structures below ?



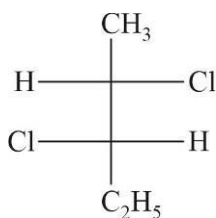
- (A) Structural isomers (B) Diastereomers
(C) Enantiomers (D) Same compound

3. Determine the number of stereoisomers in the given compound.



- (A) 2 (B) 4 (C) 8 (D) 16

4. The absolute configuration of the following compound is :



- (A) 3S, 3R (B) 2S, 3S (C) 2R, 3S (D) 2R, 3R

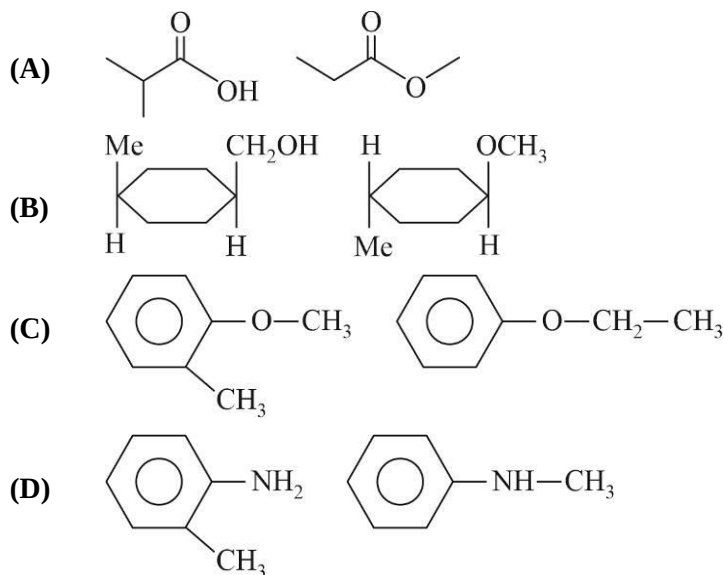
5. Consider the following three halides.

- (1) CH_3-CH_2-Cl
(2) $CH_2=CH-Cl$
(3) C_6H_5-Cl

Arrange C-Cl bond length of these compounds in decreasing order :

- (A) $1 > 2 > 3$ (B) $1 > 3 > 2$ (C) $3 > 2 > 1$ (D) $2 > 3 > 1$

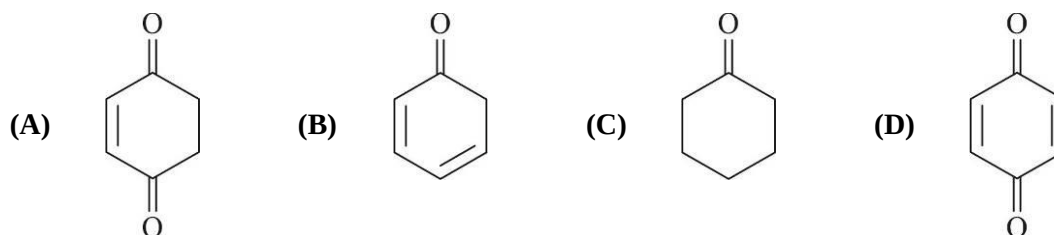
6. Which among the following is not functional isomers?



7. The correct stability order for conformation of ethylene glycol is

- (A) Anti > Gauche > Partially Eclipsed > Eclipsed
 (B) Gauche > Anti > Partially Eclipsed > Eclipsed
 (C) Eclipsed > Partially Eclipsed > Gauche > Anti
 (D) Partially Eclipsed > Eclipsed > Gauche > Anti

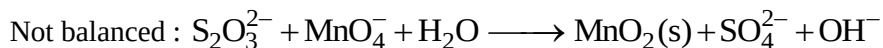
8. Which of the following compound do not undergo enolisation ?



9. Which of the following acts as an oxidising as well as reducing agent ?

- (A) Na_2O (B) H_2SO_4 (C) HNO_3 (D) HNO_2

10. 0.1 M KMnO_4 is used for following titration. What volume of the solution (in mL) will be required to react with 0.158 g of $\text{Na}_2\text{S}_2\text{O}_3$? (Molar mass of $\text{Na}_2\text{S}_2\text{O}_3$ is 158 g mol^{-1})



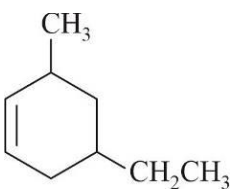
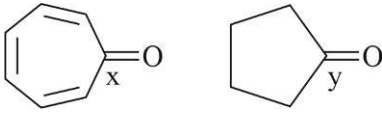
- (A) 26.7 mL (B) 50 mL (C) 65 mL (D) 75 mL

11. The maximum angle strain is seen in:

- (A) Cyclopentane (B) Cyclobutane
 (C) Cyclopropane (D) Cyclohexane

12. The IUPAC name of $\text{CH}_3 - \text{CH}_2 - \text{NH} - \text{CH}_3$ is :

- (A) Methyl ethyl amine (B) 1-methyl amino ethane
 (C) N-methyl ethanamine (D) N-ethyl methanamine

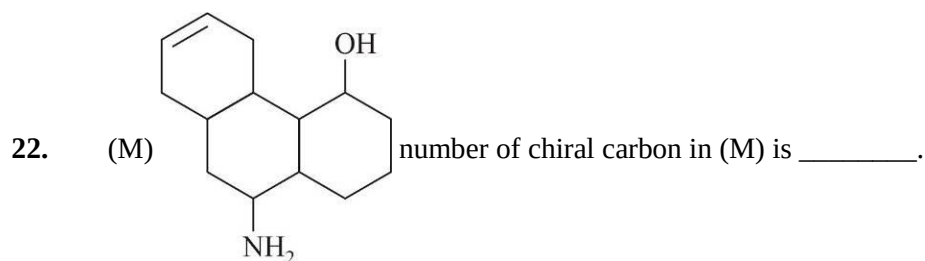
13. The IUPAC name of  is :
- (A) 1-methyl-5-ethyl cyclohex-2-ene (B) 5-ethyl-3-methyl cyclohex-1-ene
(C) 4-ethyl-6-methyl cyclohex-1-ene (D) 1-ethyl-5-methyl cyclohex-3-ene
14. Which of the following chemical reactions depicts the oxidizing behaviour of H_2SO_4 ?
- (A) $2\text{HI} + \text{H}_2\text{SO}_4 \longrightarrow \text{I}_2 + \text{SO}_2 + 2\text{H}_2\text{O}$
(B) $\text{Ba}(\text{OH})_2 + \text{H}_2\text{SO}_4 \longrightarrow \text{BaSO}_4 + 2\text{H}_2\text{O}$
(C) $\text{KCl} + \text{H}_2\text{SO}_4 \longrightarrow \text{KHSO}_4 + \text{HCl}$
(D) $2\text{PCl}_5 + \text{H}_2\text{SO}_4 \longrightarrow 2\text{POCl}_3 + 2\text{HCl} + \text{SO}_2\text{Cl}_2$
15. Which is the equivalent weight of FeSO_4 in the following reaction ?
- $$\text{FeSO}_4 \longrightarrow \text{Fe}_2(\text{SO}_4)_3$$
- (A) $\frac{M^\circ}{2}$ (B) $2M^\circ$ (C) M° (D) $\frac{M^\circ}{4}$
(M° = molecular weight of FeSO_4)
16. The correct order of bond length is:
- 
- (A) $x = y$ (B) $x > y$ (C) $x < y$ (D) None of these
17. How many esters are possible with molecular formula $\text{C}_4\text{H}_8\text{O}_2$?
- (A) 2 (B) 3 (C) 4 (D) 5
18. In NH_4NO_3 , oxidation state of both 'N' are :
- (A) -3 and $+5$ (B) $+5$ and $+5$ (C) -3 and -3 (D) $+3$ and $+5$
19. 1 mole of equimolar mixture of $\text{Fe}_2(\text{C}_2\text{O}_4)_3$ and FeC_2O_4 requires X moles of KMnO_4 in acid medium for complete reaction. The value of $10x$ is :
- (A) 90 (B) 9 (C) 10 (D) 6
20. To a 25 mL of H_2O_2 solution, excess of acidified solution of potassium iodide was added. The iodine liberated required 20 mL of 0.3 N sodium thiosulphate solution. Calculate the volume strength of H_2O_2 solution.
- (A) 0.24 (B) 1.34 (C) 0.612 (D) 2.51

SPACE FOR ROUGH WORK

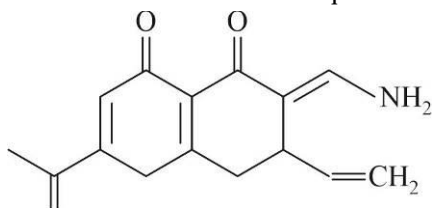
SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 99 (both inclusive).

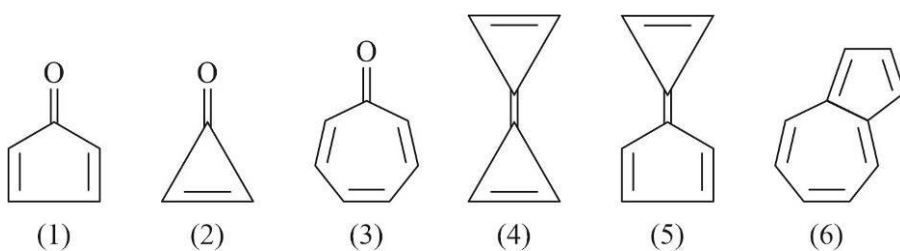
21. The maximum number of structural isomers of hexane is _____.



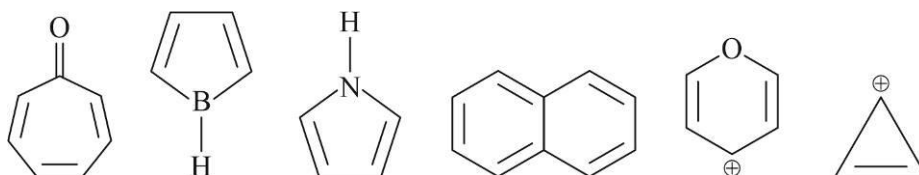
23. Find out the double bond equivalent (DBE) value of the given compound:



24. Find out total number of compounds which also have important charged contributor(resonating structure)..



25. Find out total number of compounds which are aromatic.



SPACE FOR ROUGH WORK

PART - III : MATHEMATICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- Let $a_n = \frac{4n + \sqrt{4n^2 - 1}}{\sqrt{2n+1} + \sqrt{2n-1}}$, then $\sum_{n=1}^{144} a_n$ equals:
 (A) 2465 (B) 2645 (C) 2456 (D) 2546
- For $x \in (1, \infty)$, if $P = x^3 - \frac{1}{x^3}$ and $Q = x - \frac{1}{x}$, then the minimum value of $\frac{P}{Q^2}$ equals:
 (A) $2\sqrt{3}$ (B) $4\sqrt{3}$ (C) $\sqrt{3}$ (D) $3\sqrt{3}$
- Let $2p + 3q + 4r = 15$ and if the expression $p^3 q^5 r^7$ takes the maximum value, then $p : q : r$ is equal to (where $p, q, r \in \mathbb{R}^+$):
 (A) 18 : 20 : 21 (B) 3 : 5 : 7 (C) 2 : 3 : 4 (D) 6 : 15 : 28
- The average of the numbers $n \sin n^\circ$ for $n = 2, 4, 6, \dots, 180$:
 (A) $\frac{90}{89} \cot 1^\circ$ (B) $\cot 1^\circ$ (C) $\tan 1^\circ$ (D) $90 \cot 1^\circ$
- If a and b are real numbers such that $a^2 + ab < 0$, then the expression:

$$\frac{1}{2\sqrt{-a^2 - ab}} \left(\frac{\frac{\sqrt{-a^2 - ab}}{\cos^2 x}}{a + \left(\sqrt{-a^2 - ab}\right) \tan x} - \frac{\frac{\sqrt{-a^2 - ab}}{\cos^2 x}}{-a + \left(\sqrt{-a^2 - ab}\right) \tan x} \right)$$
 simplifies to:
 (A) $\frac{1}{a + b \cos^2 x}$ (B) $\frac{1}{a + b \sin^2 x}$
 (C) $\frac{1}{a + b \sec^2 x}$ (D) Some constant independent of x
- Let z_r ($1 \leq r \leq 4$) be complex numbers such that $|z_r| = \sqrt{r+1}$ and $|30z_1 + 20z_2 + 15z_3 + 12z_4| = k |z_1 z_2 z_3 + z_2 z_3 z_4 + z_3 z_4 z_1 + z_4 z_1 z_2|$. Then the value of k equals:
 (A) $|z_1 z_2 z_3|$ (B) $|z_2 z_3 z_4|$ (C) $|z_3 z_4 z_1|$ (D) $|z_4 z_1 z_2|$
- If $1, \alpha_1, \alpha_2, \dots, \alpha_{2008}$ are $(2009)^{\text{th}}$ roots of unity, then the value of $\sum_{r=1}^{2008} r(\alpha_r + \alpha_{2009-r})$ equals:
 (A) 2009 (B) 2008 (C) 0 (D) -2009
- When the polynomial $5x^3 + Mx + N$ is divided by $x^2 + x + 1$ the remainder is 0. The value of $(M + N)$ is equal to:
 (A) -3 (B) 5 (C) -5 (D) 15

9. There are 100 different books in a shelf. Number of ways in which 3 books can be selected so that no two of which are neighbours is:
 (A) ${}^{100}C_3 - 98$ (B) ${}^{97}C_3$ (C) ${}^{96}C_3$ (D) ${}^{98}C_3$
10. There are six periods in each working day of a school. Number of ways in which 5 subjects can be arranged if each subject is allotted at least one period and no period remains vacant is:
 (A) 210 (B) 1800 (C) 360 (D) 120
11. Number of rectangles in the grid shown which are not squares is: (each smallest unit is a square)
- | | | | | | |
|--|--|--|--|--|--|
| | | | | | |
| | | | | | |
| | | | | | |
| | | | | | |
- (A) 160 (B) 162 (C) 170 (D) 185
12. If $g(x) = \left(4\cos^4 x - 2\cos 2x - \frac{1}{2}\cos 4x - x^7\right)^{1/7}$, then the value of $g(g(100))$ is equal to:
 (A) -1 (B) 0 (C) 1 (D) 100
13. Let $f(x) = (x+2)e^{\ln(x+2)}$ and $g(x) = \frac{e^{\left(\frac{1}{-\log_x e}\right)}}{x} - \frac{2x}{e^{-\ln 2}}$. If $h(x) = f(x) + g(x)$, then the smallest positive integer in the range of $h(x)$ is:
 (A) 2 (B) 3 (C) 6 (D) 7
14. Let $f: \mathbb{R} \rightarrow [1, \infty)$ be defined by $f(x) = x^2 - 6ax + 3 - 2a + 9a^2$. The integral value of a for which $f(x)$ is surjective, is equal to:
 (A) 0 (B) 1 (C) 2 (D) -3
15. Minimum distance between the curves $y^2 = x - 1$ and $x^2 = y - 1$ is equal to:
 (A) $\frac{3\sqrt{2}}{4}$ (B) $\frac{5\sqrt{2}}{4}$ (C) $\frac{7\sqrt{2}}{4}$ (D) $\frac{\sqrt{2}}{4}$
16. TP and TQ are tangents to the parabola, $y^2 = 4ax$ at P and Q. If the chord PQ passes through the fixed point $(-a, b)$, then the locus of T is:
 (A) $ay = 2b(x - b)$ (B) $bx = 2a(y - a)$ (C) $by = 2a(x - a)$ (D) $ax = 2b(y - b)$
17. If the tangents and normals at the extremities of a focal chord of a parabola intersect at (x_1, y_1) and (x_2, y_2) respectively, then:
 (A) $x_1 = x_2$ (B) $x_1 = y_2$ (C) $y_1 = y_2$ (D) $x_2 = y_1$
18. A circle has the same centre as an ellipse and passes through the foci F_1 and F_2 of the ellipse, such that the two curves intersect in 4 points. Let P be any one of their point of intersection. If the major axis of the ellipse is 17 and the area of the triangle PF_1F_2 is 30, then the distance between the foci is:
 (A) 11 (B) 12 (C) 13 (D) 14

19. The eccentricity of the ellipse $(x-3)^2 + (y-4)^2 = \frac{y^2}{9}$ is:
 (A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{3\sqrt{2}}$ (D) $\frac{1}{\sqrt{3}}$
20. If P is any point on ellipse with foci S_1 and S_2 and eccentricity is $\frac{1}{2}$ such that $\angle PS_1S_2 = \alpha, \angle PS_2S_1 = \beta, \angle S_1PS_2 = \gamma$, then $\cot \frac{\alpha}{2}, \cot \frac{\gamma}{2}, \cot \frac{\beta}{2}$ are in:
 (A) A.P. (B) G.P.
 (C) H.P. (D) Not A.P., G.P. & H.P.

SPACE FOR ROUGH WORK

SECTION-2

This section contains Five (05) Numerical Value Type Questions. The answer to each question is an integer ranging from 0 to 99 (both inclusive).

21. Let circles C_1 and C_2 on Argand plane be given by $|z+1|=3$ and $|z-2|=7$ respectively. If a variable circle $|z-z_0|=r$ be inside circle C_2 such that it touches C_1 externally and C_2 internally then locus of z_0 describes a conic E. If eccentricity of E can be written in simplest form as $\frac{p}{q}$ where $p, q \in \mathbb{N}$, then the value of $(p+q)$, is _____.
22. If $f: \mathbb{R} \rightarrow \mathbb{R}$ be an injective mapping and p, q, r are non-zero distinct real quantities satisfying $f\left(\frac{p}{r}\right) = f\left(\frac{p-q}{q-r}\right)$ and $f\left(\frac{q}{r}\right) = f\left(\frac{r}{p}\right)$. If the graph of $g(x) = px^2 + qx + r$ passes through $M(1, 6)$, then the value of q is _____.
23. If the sum of all values of x satisfying the system of equations:
 $\tan x + \tan y + \tan x \cdot \tan y = 5$
 $\sin(x+y) = 4 \cos x \cdot \cos y$
 is $\frac{k\pi}{2}$, where $x \in \left(0, \frac{\pi}{2}\right)$ then the value of k , is _____.
24. Let $S = \frac{1}{\sin 8^\circ} + \frac{1}{\sin 16^\circ} + \frac{1}{\sin 32^\circ} + \dots + \frac{1}{\sin 4096^\circ} + \frac{1}{\sin 8192^\circ}$. If $S = \frac{1}{\sin \alpha^\circ}$ where $\alpha \in (0, 90^\circ)$, then the value of α is _____.
25. If x, y, z are arbitrary positive real numbers satisfying the equation $4xy + 6yz + 8zx = 9$ and maximum possible value of (xyz) is M , then, $(256 M)$ is:

SPACE FOR ROUGH WORK

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